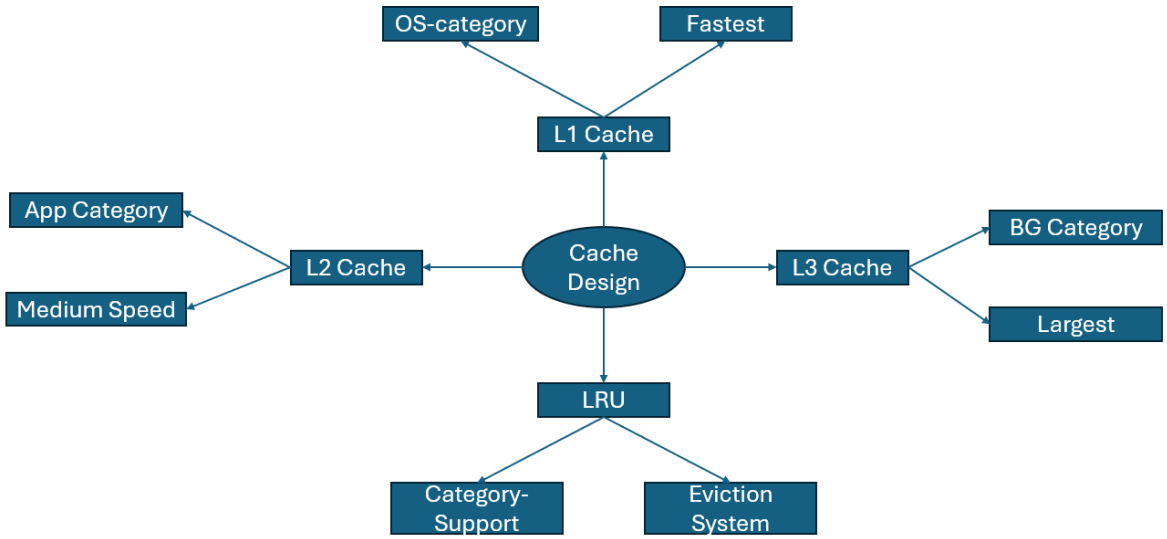
CACHE MEMORY DESIGN

Contents

Introduction

This Cache Memory Architecture is designed to help improve the performance of computers. It is also made as an extension of the RAM Memory Architecture design. This cache design was created because the CPU fetches the most frequently used instructions. The most frequently used instructions are stored in cache as it is directly in the CPU and has limited storage space. This cache design mainly uses pre-existing systems like LRU and the current levels. This design uses the category-based memory only since there is no need for encrypting cache. This documentation will explain the features and trade-offs of this cache design.

This cache design, like the RAM memory architecture cannot promise perfect performance. Instead, this cache design can promise to further improve performance alongside the RAM memory architecture. This cache design may take years to implement along with the RAM memory architecture. Engineers and OS vendors may need to work together to implement this cache design. This cache design's categories can be adapted to fit specific computing systems.



The spider diagram above shows a layout of the Cache Memory Architecture. For those that aren't familiar with computing architecture, LRU stands for least recently used. The same 3 categories as the RAM memory architecture are used here since cache and RAM perform the same task. This task is storing instructions which the CPU fetches, decodes, and then executes.

Cache Category-Based Memory

- **Cache category-based memory is where instructions within cache are sorted by categories instead of sharing levels.**
- **This cache category-based memory uses the two pre-existing systems of LRU and L1, L2 and L3 cache.**
- **For L1 Cache, it should store the most frequent OS category instructions.**
- **This is because L1 cache is the smallest but also the fastest.**
- **This is important because OS instructions need instant access with no waiting.**
- **This benefits performance as the operating system is usually the most critical in most computer systems.**
- **For L2 Cache, it should store the most frequent App category instructions.**
- **This is because L2 cache has medium size and medium speed compared to L1 or L3.**
- **This is important because application instructions need quick but not instant access compared to the operating system.**
- **This benefits performance as applications and the operating system do not have to compete for space and access.**
- **However, this means that applications may run slower compared to the operating system which could impact gaming performance.**
- **For L3 Cache, it should store the most frequent background category instructions.**
- **This is because L3 cache is the biggest but also the slowest.**
- **This is important because background processes do not require as much speed compared to applications and OS processes.**
- **This benefits performance as background processes do not slow down application and operating system processes.**
- **However, this also means that certain background processes like anti-virus or certain security software may be slower.**
- **This cache design uses LRU for its eviction policy.**
- **LRU must be configured for this design to be applied within each category and not globally.**
- **LRU would evict the least frequently used OS data in L1 cache.**
- **LRU would evict the least frequently used App data in L2 cache**
- **LRU would evict the least frequently used BG data in L3 cache**
- **This is important to ensure that the cache can stay updated with the most frequently used instructions.**
- **The categories in each level can be adapted based on the system.**
- **For example, in gaming systems L1 cache may use the app category.**

Cache Design Trade-offs

- **This cache design has trade-offs just like any other system I have built.**
- **There are 3 main trade-offs I've identified.**
- **The first trade-off is Heat.**
- **This trade-off exists because the CPU can access specific instructions from each level of cache leading to faster fetching meaning more instructions executed quickly.**
- **This means that the CPU will heat up as heat is produced as more work is done by the CPU.**
- **This trade-off can be managed through cooling systems and limiting clock-speed for the CPU.**
- **The second trade-off is Application/Background Performance.**
- **This trade-off exists for two reasons.**
- **The first reason is that application instructions are stored in L2 cache which is medium speed and size.**
- **This means that some applications may run slower compared to operating system processes.**
- **The second reason is that background process instructions are stored in L3 cache which is the largest but slowest.**
- **This means that some background processes like updates and security software may have slower performance.**
- **This trade-off can be managed by adapting categories per level based on use case.**
- **For example, Applications may be in L1 cache for gaming systems.**
- **The third trade-off is cache misses.**
- **This trade-off exists because sometimes the instructions that the CPU is looking for are not in cache.**
- **This means the CPU searches all levels of cache before going to RAM which can be slow and time-consuming.**
- **This trade-off can be managed through a system where if the instruction is not stored in a specific level, then the CPU immediately heads to RAM instead of searching the other levels of cache.**
- **This improves performance because it saves time and allows the CPU to immediately fetch the instruction from RAM.**
- **This is also supported by category-based memory in RAM as the CPU can still fetch instructions quickly.**